香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen
人工智能學院
School of Artificial Intelligence

AIE1901-S1 Lifelong AI: Lecture 2: Machine Learning Basics

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2026/9/7

Agenda

- Inside the blackbox

A more formal definition of (supervised) machine learning

- What is machine learning generalization?

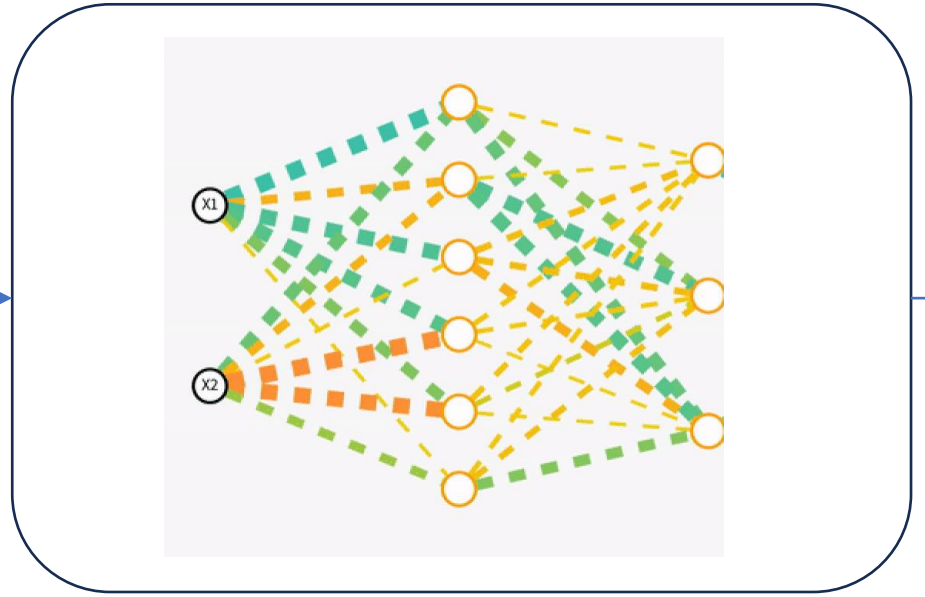
Bias-Variance tradeoff and "no free lunch theorem"

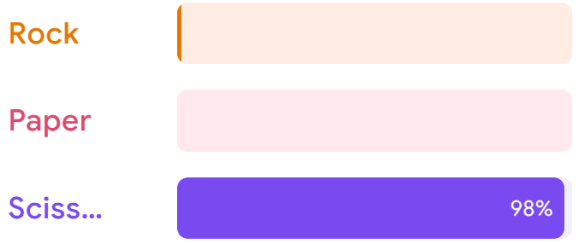
- Next: In-domain vs out-of-domain learning

Review: Neural Networks

Example: Hand gesture classification





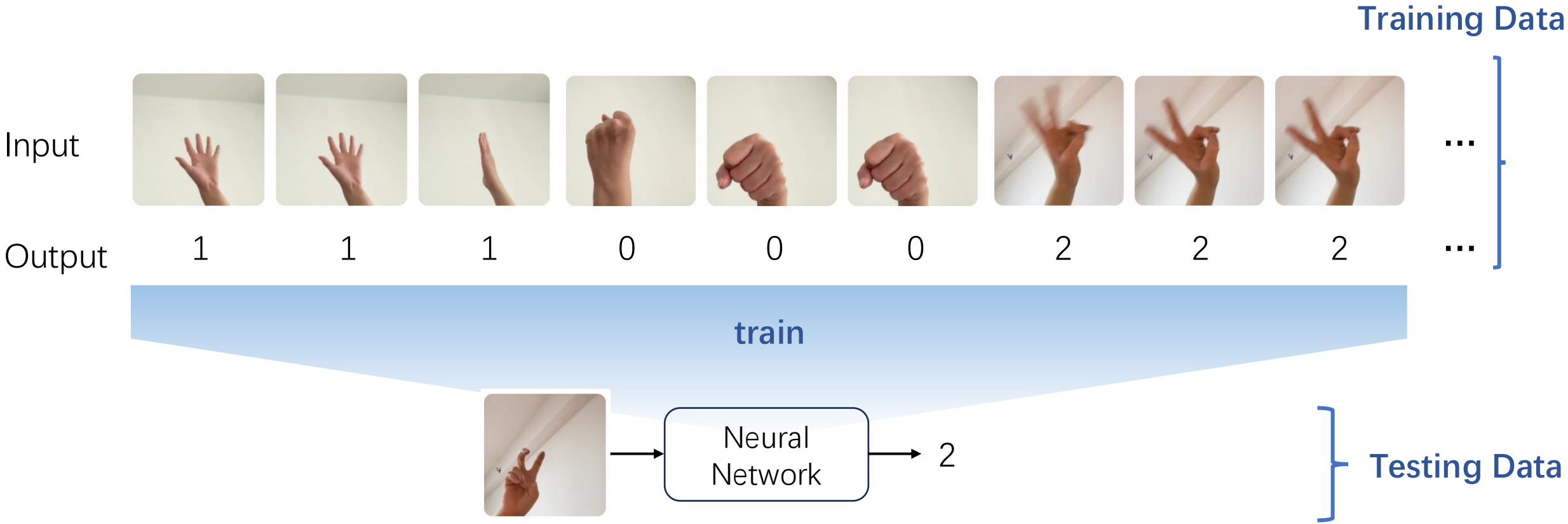


Input

Neural Network

Output

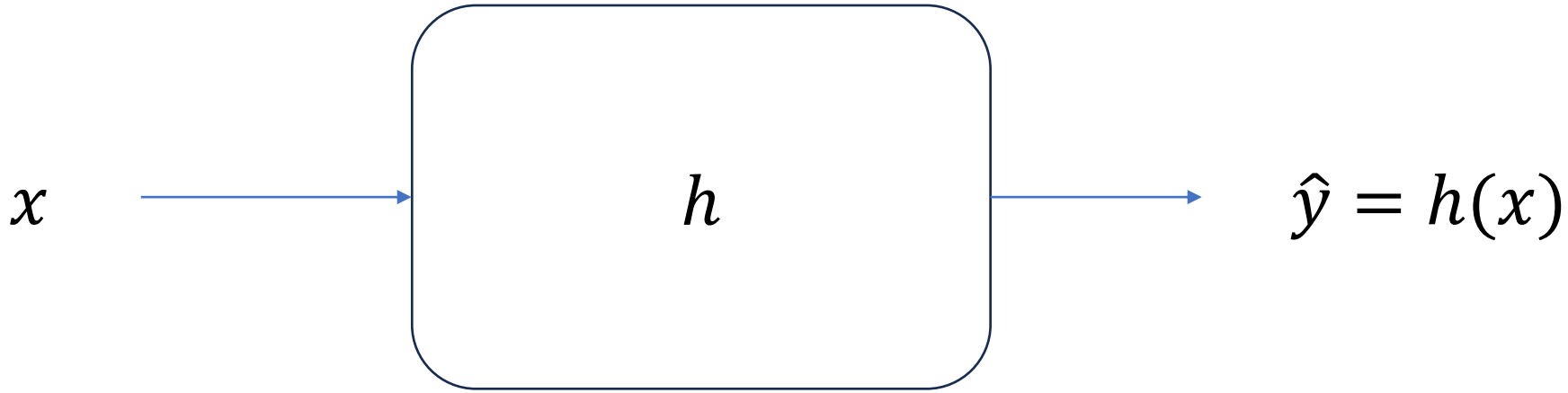
How Neural Network Learns?



Given a number of **input-output examples**, **(supervised) learning** aims to **find the optimal weights** that can map any new data to the correct output

A high-level view of supervised learning

Input space: $\mathcal{X}$, Target space : $\mathcal{Y}$
Hypothesis function: $h: \mathcal{X} \to \mathcal{Y}$



Input

Hypothesis Function

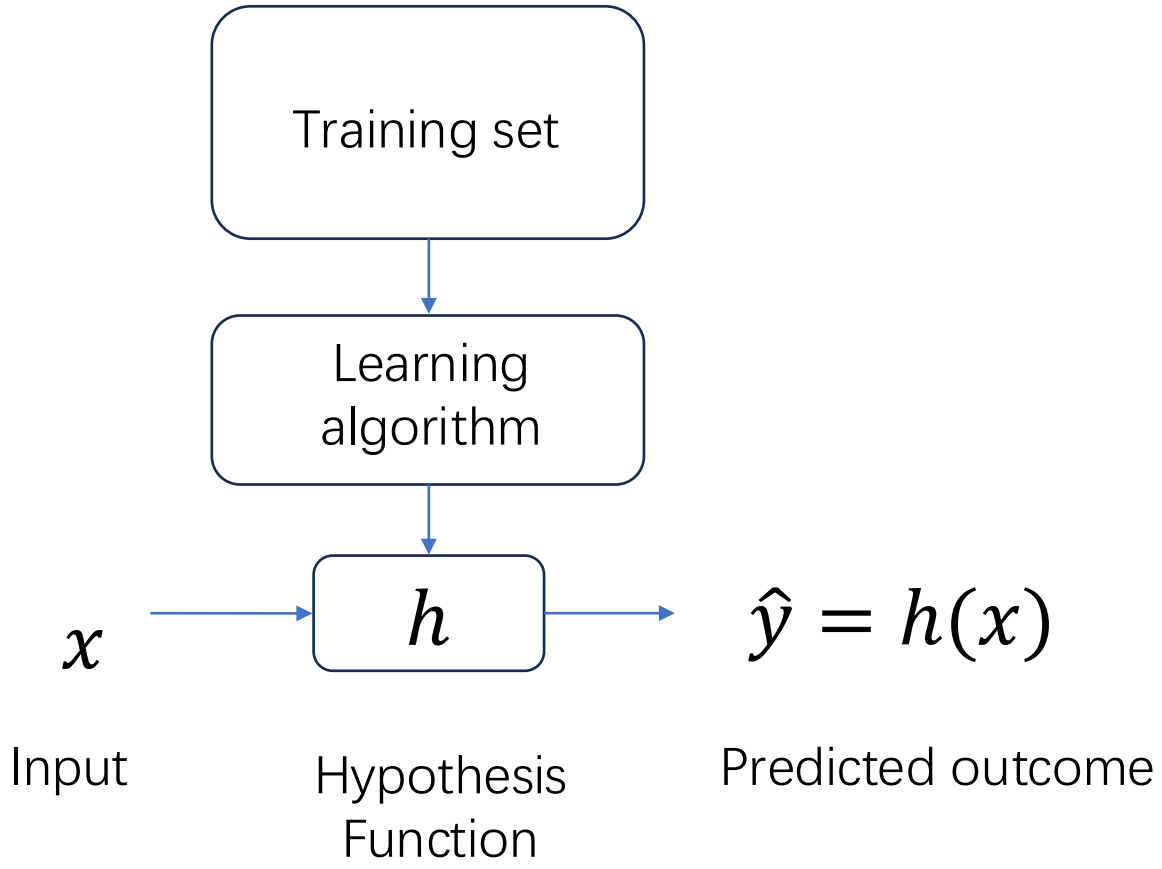
Predicted outcome

e.g. 

e.g. 2

A high-level view of supervised learning

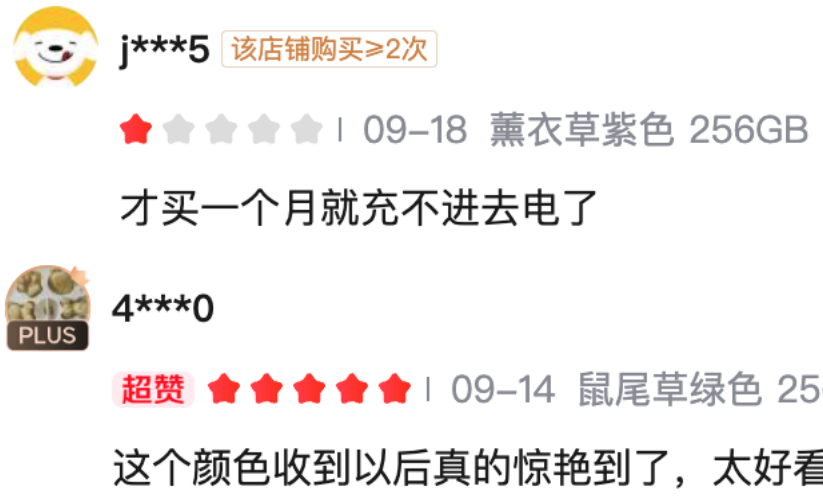
Given a **training dataset**, learn a hypothesis function $h: \mathcal{X} \rightarrow \mathcal{Y}$ so that $h(x)$ is a good predictor for the corresponding y.



Review: supervised learning

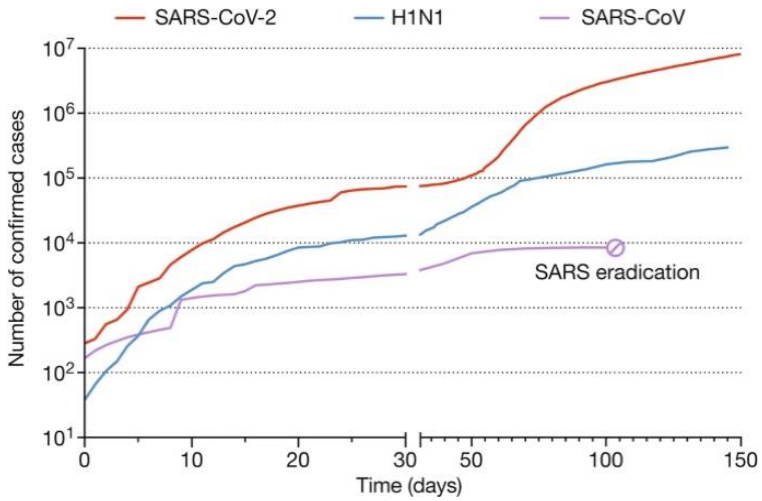
y is discrete (categorical):
classification problem





y is continuous (real value):
regression problem



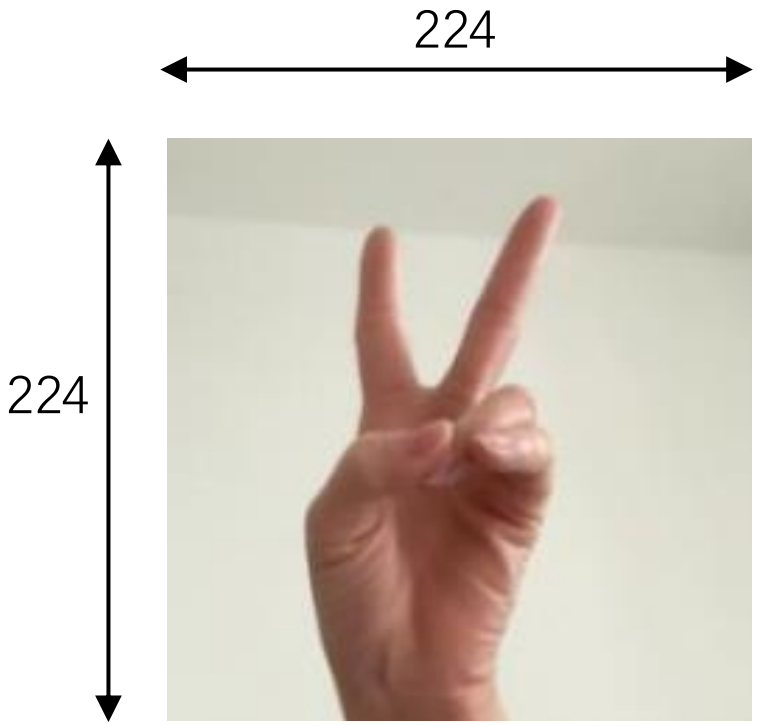


Training Dataset

A **training dataset** $D_{\text{train}} = (X, Y)$ is a collection of input and labels:

$$X = \{x^{(1)}, \dots, x^{(m)}\},$$
$$Y = \{y^{(1)}, \dots, y^{(m)}\}$$

Each input (data point) $x^{(i)}$ is represented by n **features**.



What is the dimension n of this input?

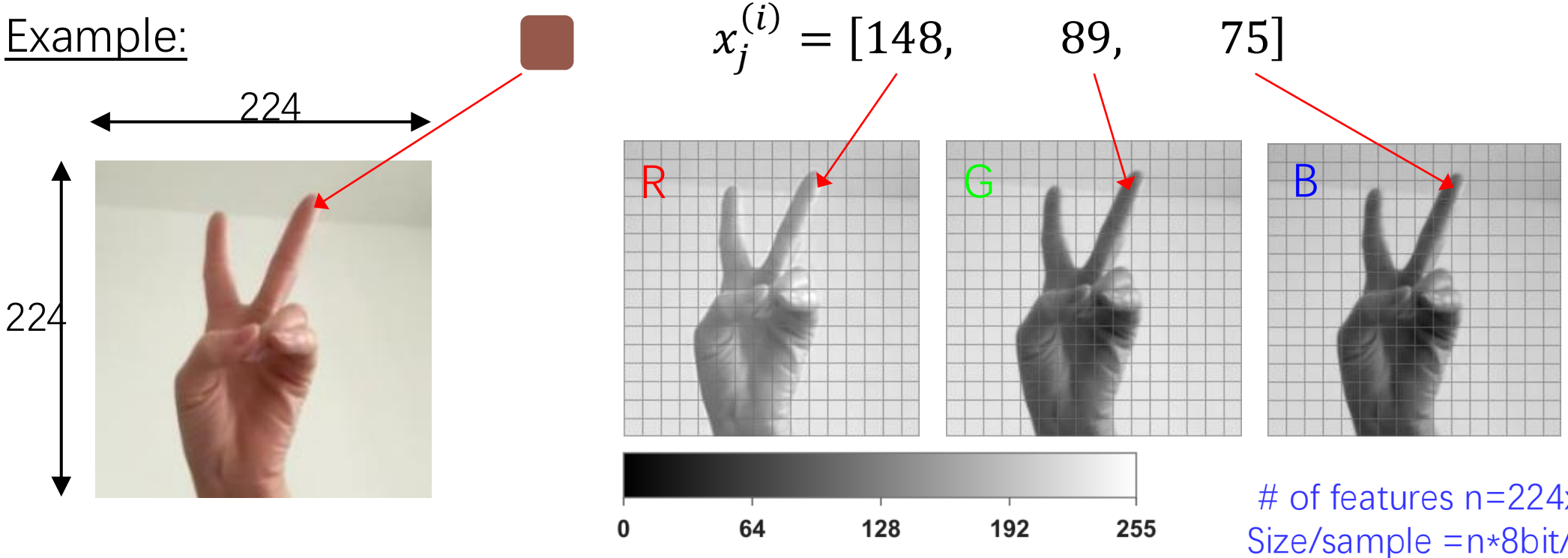
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Example:



of features n=224x224x3
Size/sample =n*8bit/feature

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For classification tasks, label y is an integer in $\{1, 2, \dots, K\}$, often encoded as a n-dim **one-hot vector**

Example: $\mathcal{Y} = \{0, 1, 2\}$

Class 0 $\rightarrow [1, 0, 0]$
Class 1 $\rightarrow [0, 1, 0]$
Class 2 $\rightarrow [0, 0, 1]$

Convention: Start numbering from 0

Measuring Learning Performance

Quantitatively evaluate the ability of a machine learning algorithm for a given task.

- **Classification accuracy** for is the fraction of correctly predicted samples in a dataset.

$$\text{Accuracy(h)} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\{y^{(i)} = h(x^{(i)})\},$$

Example:



Ground Truth y	0	2	1	1	0	2
Predicted $\hat{y} = h(x)$	2	2	0	1	0	2

Accuracy(h) = ?

Measuring Learning Performance

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$$\text{Accuracy(h)} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\left\{y^{(i)} = h\left(x^{(i)}\right)\right\},$$

- Accuracy is hard to optimize directly!
- During training, we minimize a differentiable proxy

Example: **cross-entropy loss** for classification

$$\hat{L}(h) = -\frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{K} y_k^{(i)} \log p_{h,k}\left(x^{(i)}\right).$$

Measuring Learning Performance

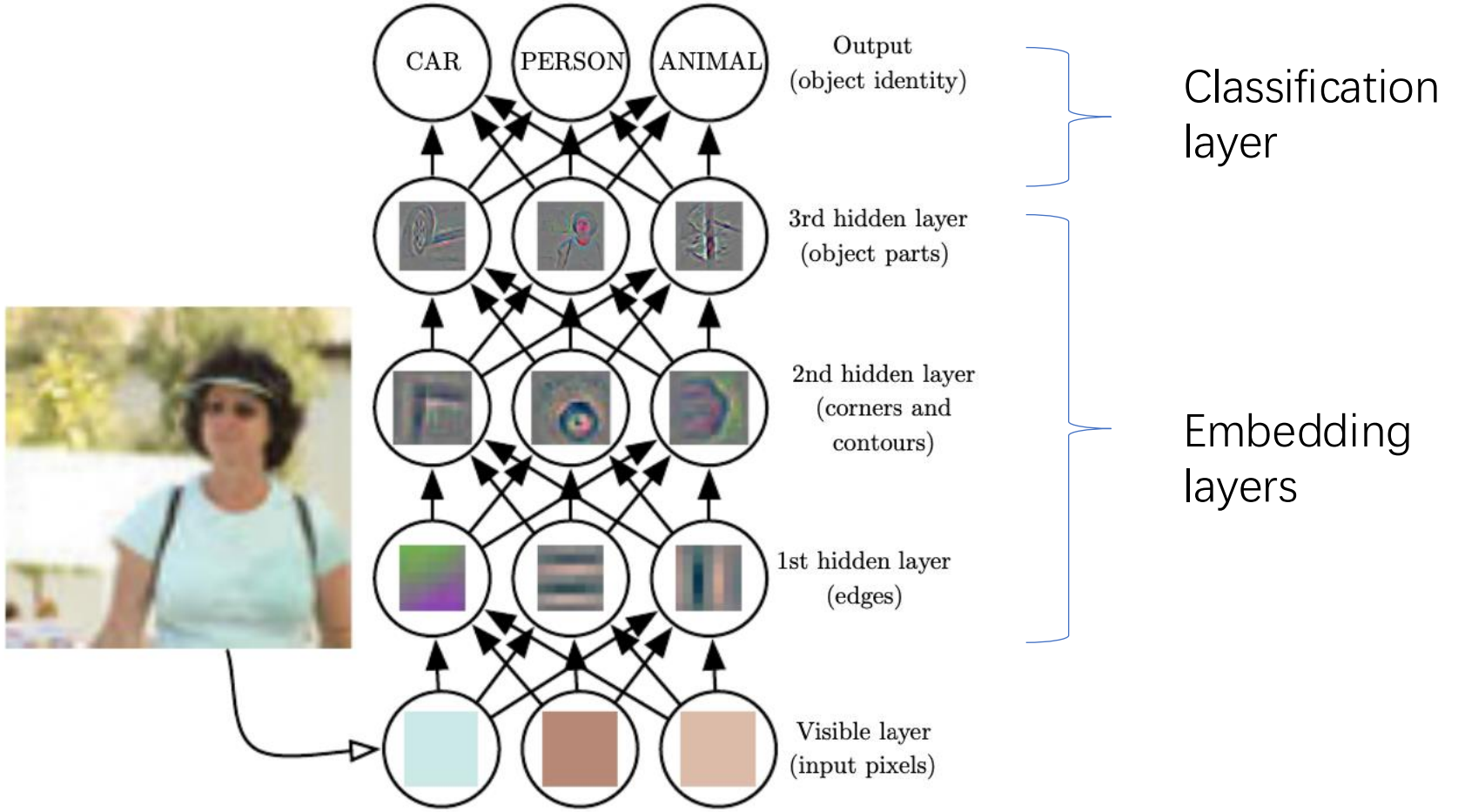
Example: **cross-entropy loss** for classification

$$L(h) = -\frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{K} y_k^{(i)} \log p_{h,k}\left(x^{(i)}\right).$$

Sample i	1	2	3	4	5	6
Ground truth $y^{(i)}$	[1,0,0]	[0,0,1]	[0,1,0]	[0,1,0]	[1,0,0]	[0,0,1]
Prediction $p_h(x)$	[0.10, 0.15, 0.75]	[0.08, 0.22, 0.7]	[0.65, 0.25, 0.10]	[0.15, 0.72, 0.13]	[0.8, 0.12, 0.08]	[0.05, 0.18, 0.77]
Predicted class	2 ✗	2 ✓	0 ✗	1 ✓	0 ✓	2 ✓
Loss $l^{(i)}$	−log 0.10	−log 0.70	−log 0.25	−log 0.72	−log 0.80	−log 0.77

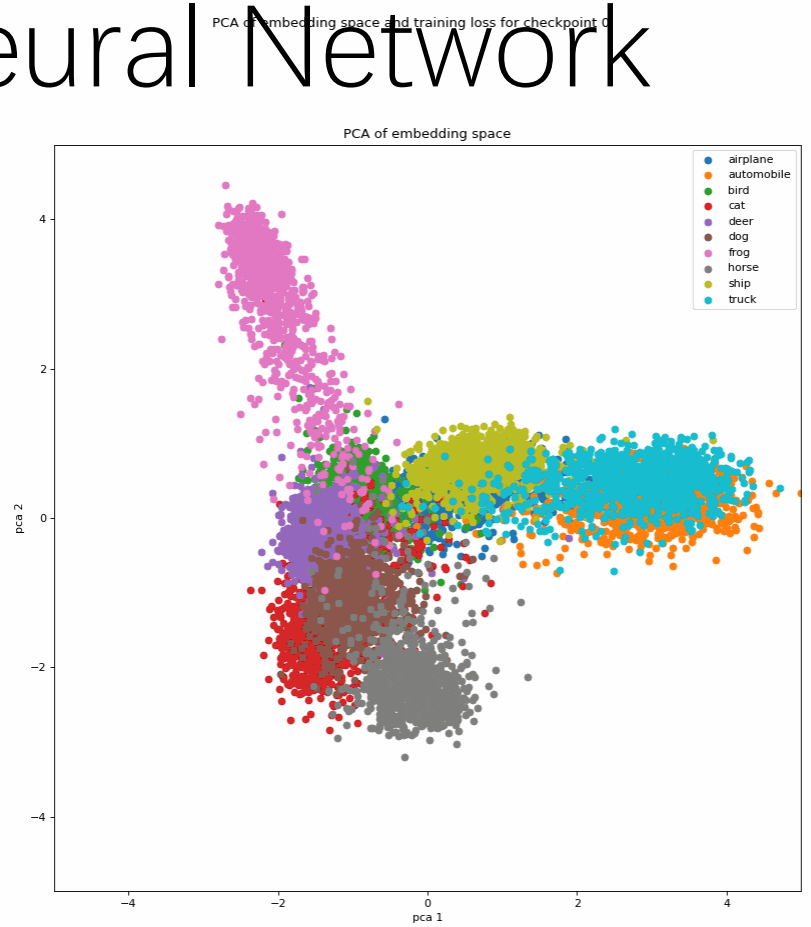
L(h) = 4.859/6 =0.810

Training a Classification Neural Network



Training a Classification Neural Network

- For any set of network weights, NN learns an **embedding function** f that each sample to an **embedding** vector e.g. $z^{(i)} = f\left(x^{(i)}\right) = [0.25, -0.41, \ldots]$
- In this new space, NN also learns a **head classifier** $g(z)$ that will estimate how likely is $x^{(i)}$ to assigned to class j
- The hypothesis function is $h(x) = g\left(f(x)\right)$
- Neural Network is often trained by minimizing loss function $L(h)$using **iterative algorithms**, i.e. update the weights in small steps until it stabilizes (converge)



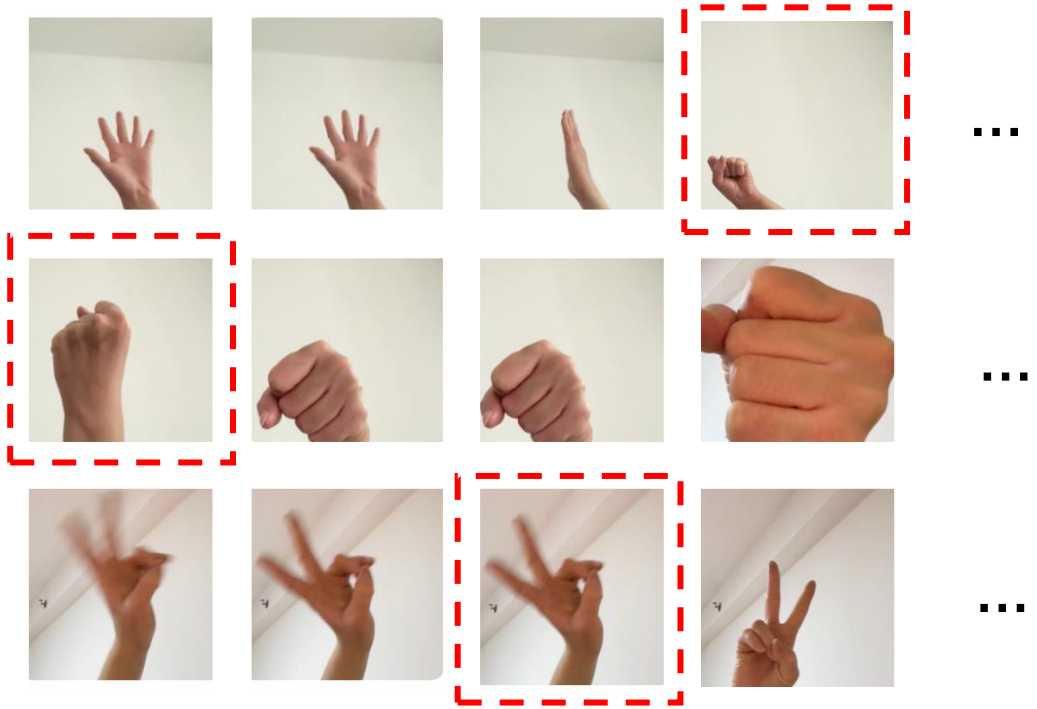


https://daniel-bethell.co.uk/posts/visualising-embeddings/
https://huggingface.co/blog/MarkusStoll/embeddings-during-fine-tuning-of-vision-transform

Training vs. Testing Performance

- To evaluate learning performance, we must test on **previously unseen input**!
- Separate **test dataset** from **training dataset** (i.e. no overlap sample $D_{train} \cap D_{test} = \emptyset$)

training set



...

...

...

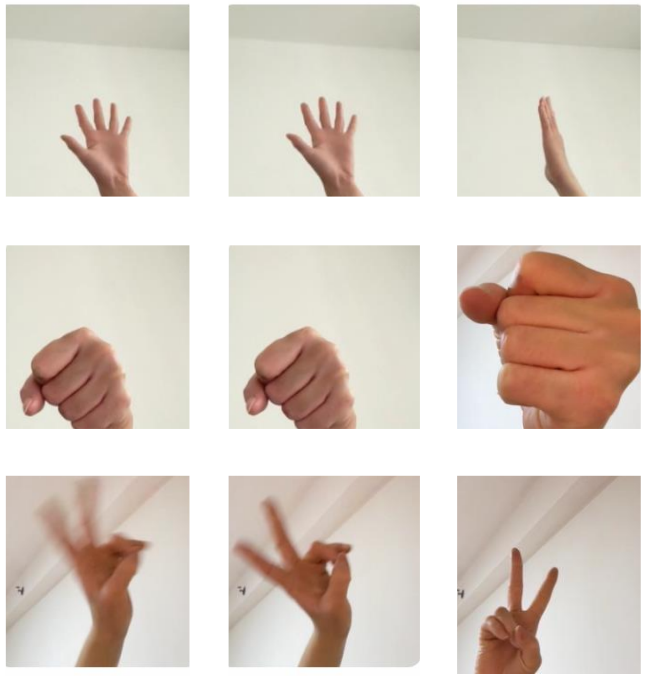
testing set

In TeachableMachine, 15% of the training data per class will be randomly split into a testing set

Training vs. Testing Performance

- To evaluate learning performance, we must test on **previously unseen input**!
- Separate **test dataset** from **training dataset** (i.e. no overlap sample $D_{train} \cap D_{test} = \emptyset$)

training set



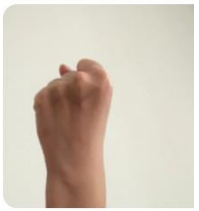
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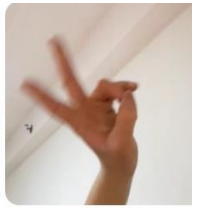
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testing set







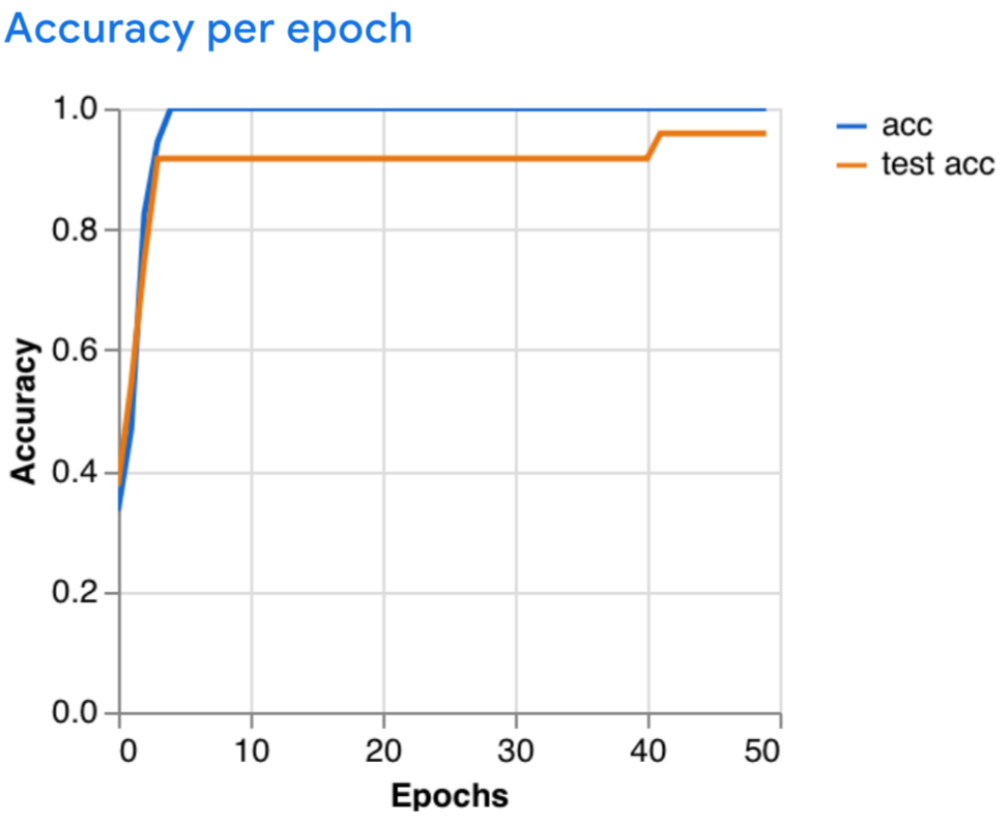
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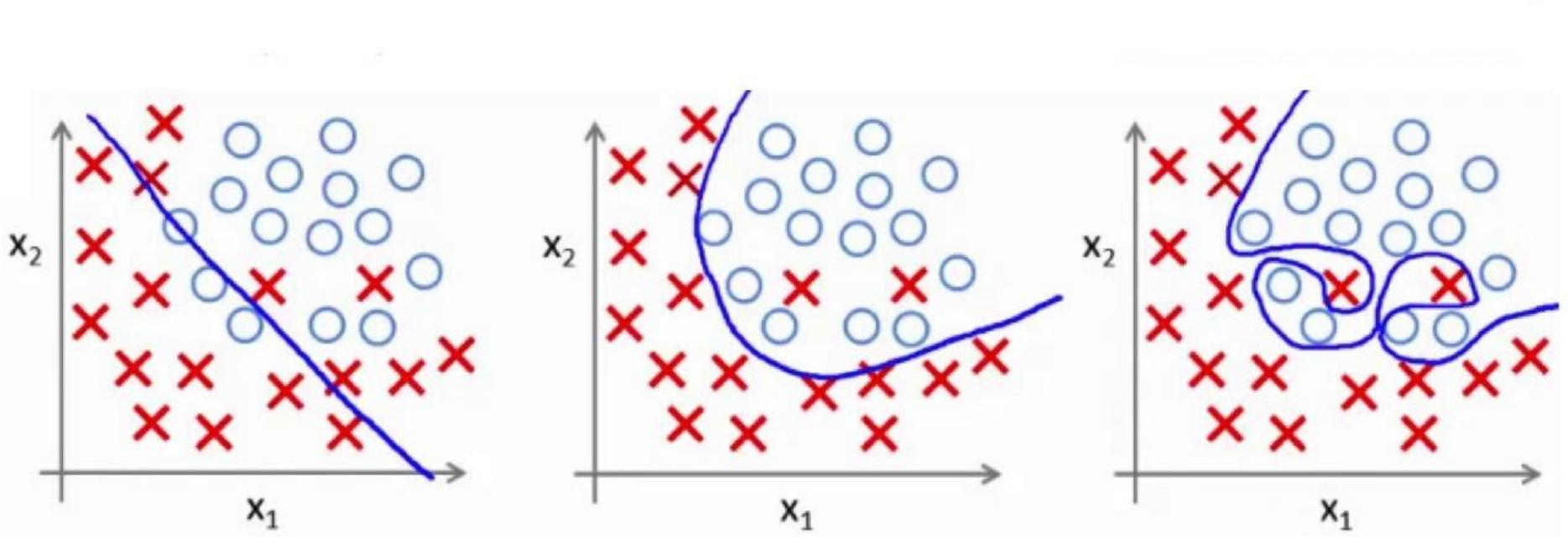
Goal of Machine Learning

- Make training error small (optimization)
- Make the gap between the training and testing error small!



Overfit & Underfit

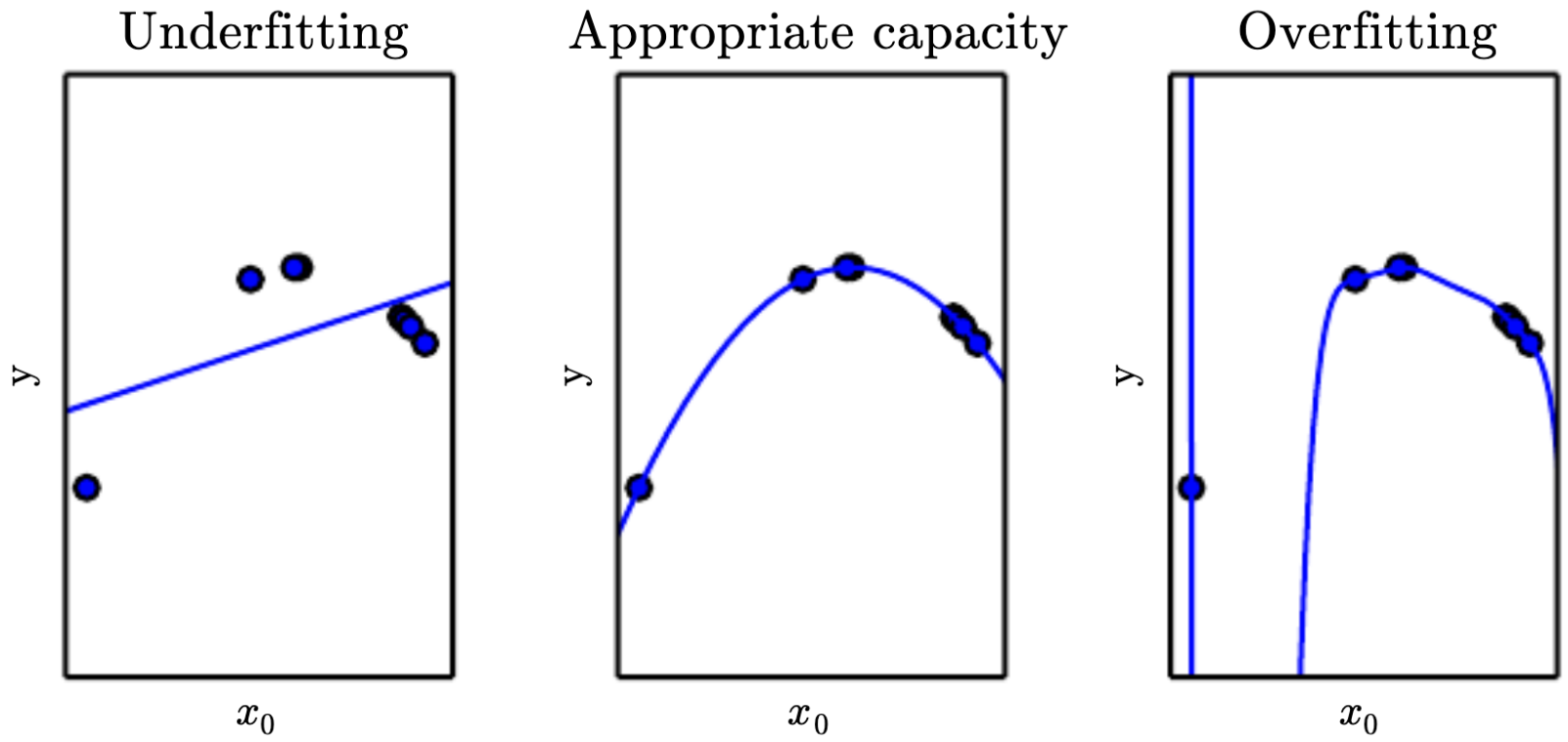
- **Underfit**: Both training error and testing error are large
- **Overfit:** Training error is small, testing error is large



underfit **overfit**

Overfit & Underfit

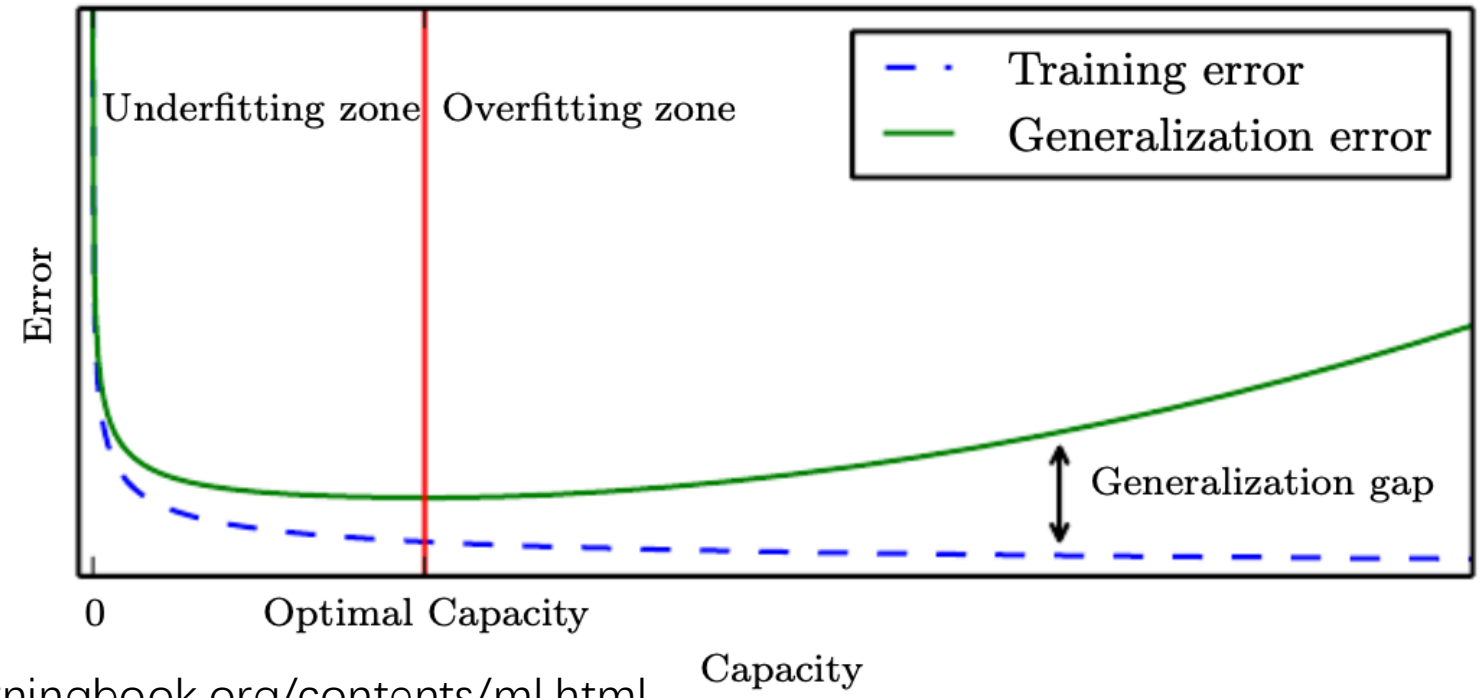
- **Underfit**: Both training error and testing error are large
- **Overfit:** Training error is small, testing error is large



Model Capacity

Changing a model's **capacity** controls whether it is more likely to overfit or underfit

- Choose a model's hypothesis space: e.g. increase # of neurons (adding parameters)
- Find the best among a family of hypothesis functions

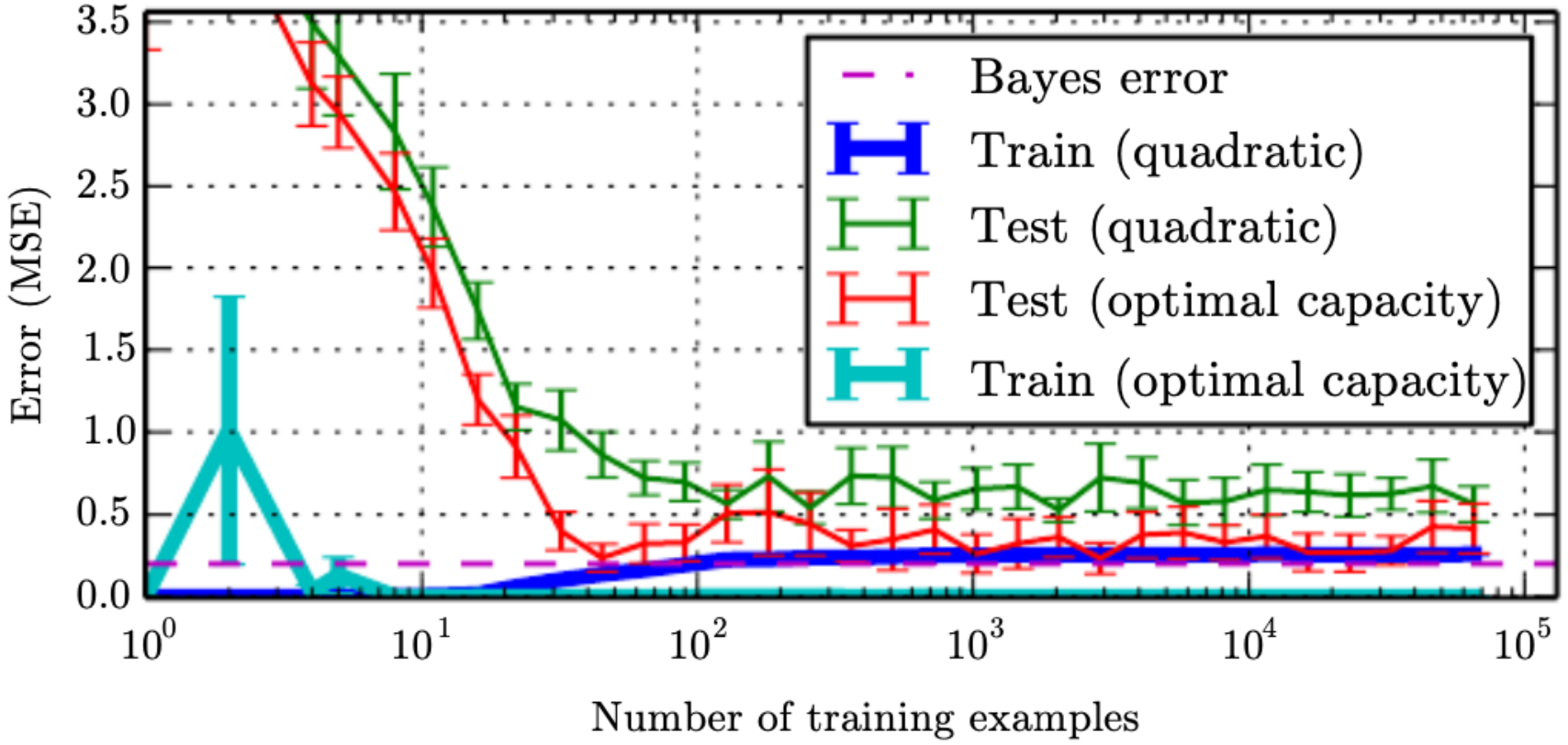


https://www.deeplearningbook.org/contents/ml.html

No Free Lunch Theorem

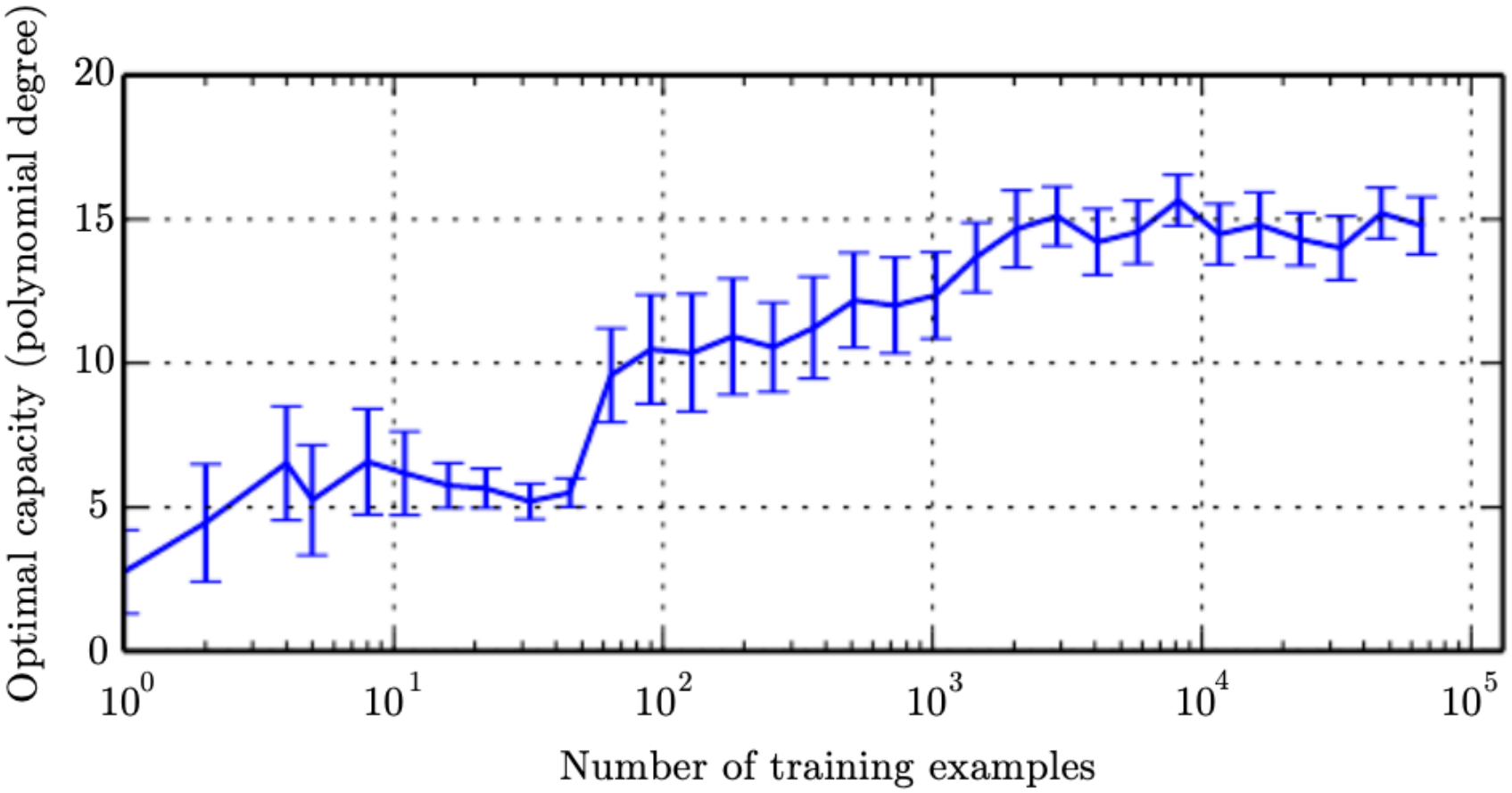
- **Learning theory**： the study to measure whether algorithm generalize well from a finite training set of examples.
  - Since we can not see future testing data, machine learning find rules that are probably correct about members of most the set
- **No free lunch theorem**: No machine learning algorithm is universally better than another (Wolpert 1996)

No Free Lunch Theorem



- **More data improves generalization:** More data reduces test error and the train–test gap.
- **More data supports more complex models:** Optimal model capacity increases with dataset size, then plateaus once sufficient capacity is reached.

No Free Lunch Theorem



The best model depends on the amount of data and the task

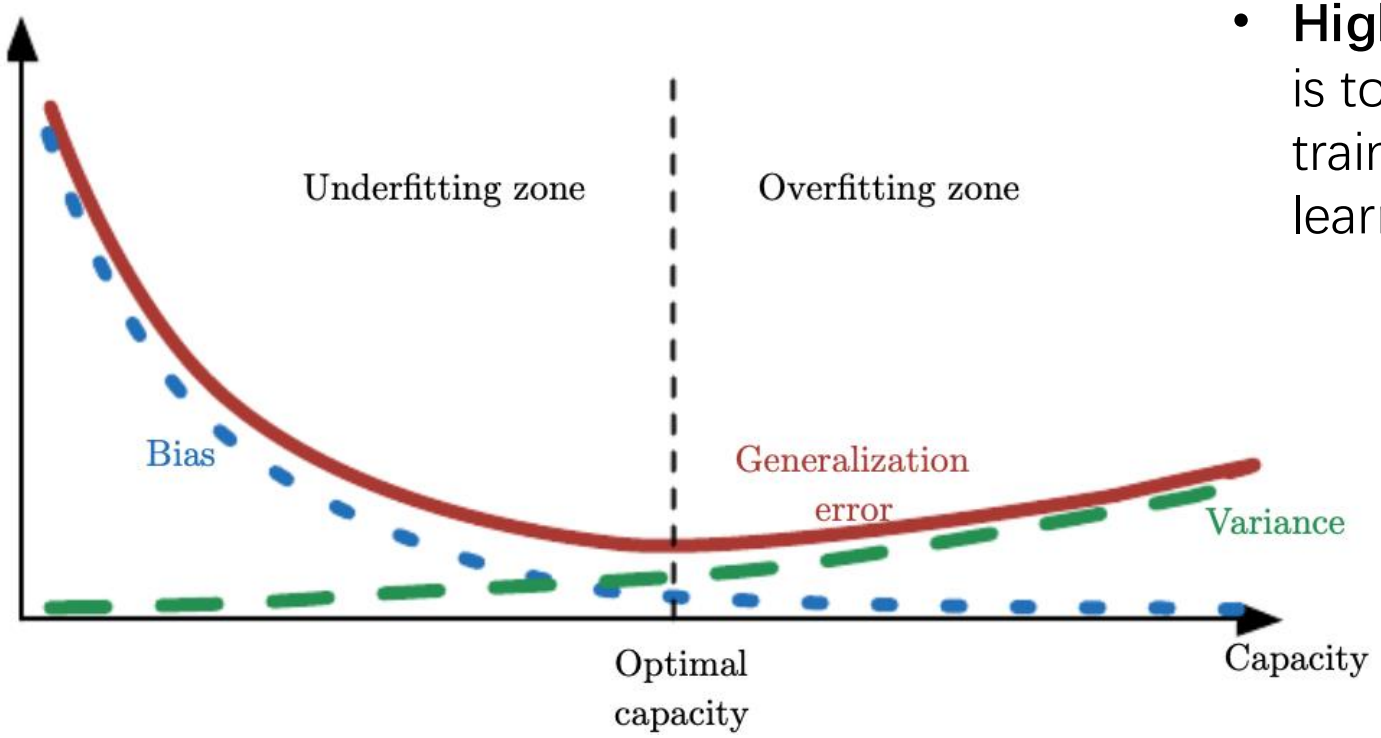
There is no universally optimal model!

Bias-Variance Trade Off

The generalization (test) error can be decomposed into two parts:

- **Bias:** Ability of the model to fit any pattern
- **Variance**: Sensitivity of the model due to choices of training sample

- **High bias:** The model is too simple and misses important patterns



- **High variance:** The model is too sensitive to its training examples and learns noise

From In-Domain to Out-of-Domain Learning

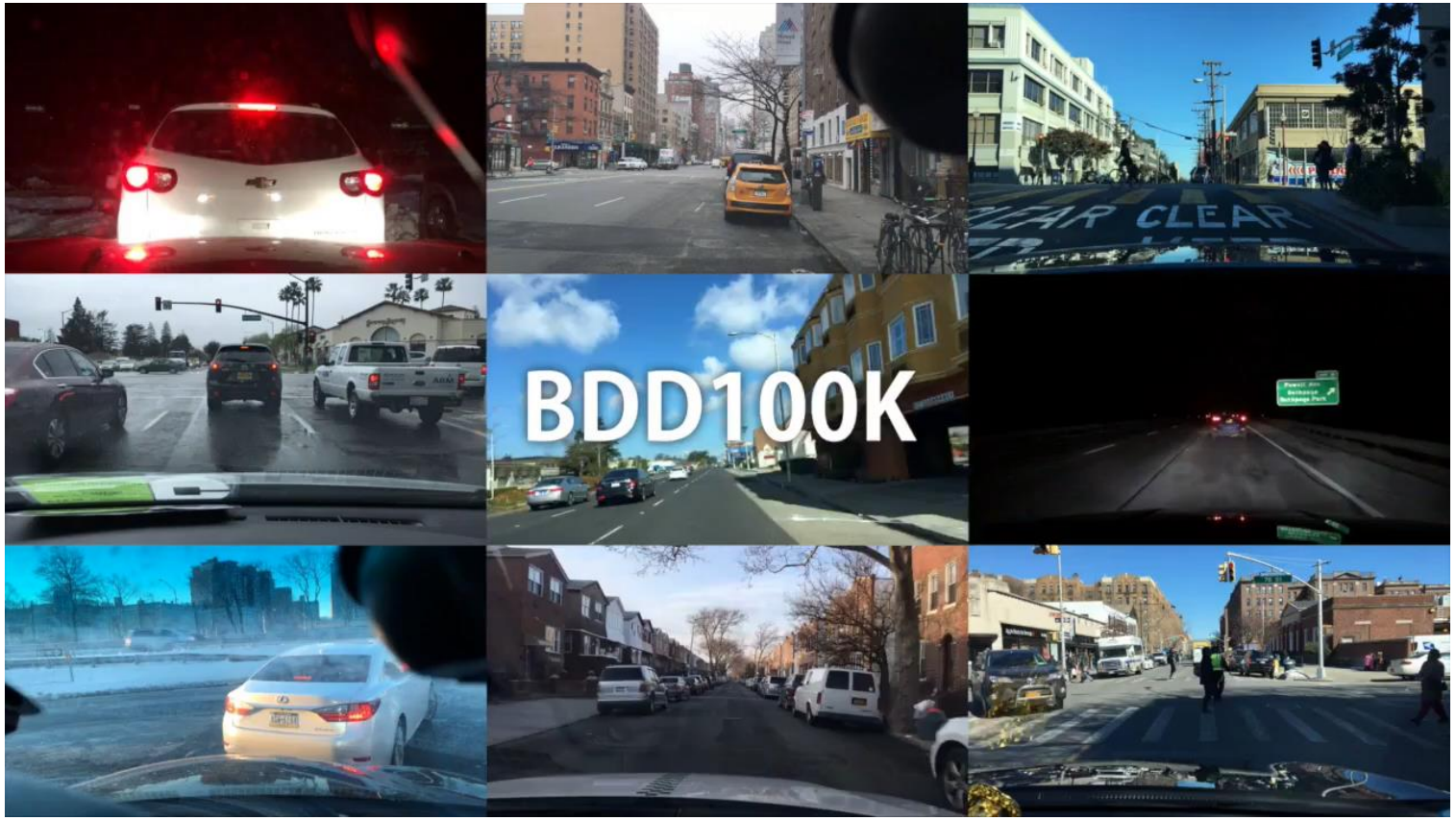
Classic learning theory assumes training and test examples are i.i.d.
(**independent** samples drawn from the **same data distribution**)

When these assumptions are not true:

- **Not independent:** Web-cam dataset may capture many similar input
- **Training and testing have different data distribution**: When there is distribution shift in real-world test data

$$P_{\text{test}}(X, Y) \neq P_{\text{train}}(X, Y)$$

From In-Domain to Out-of-Domain Learning



This is **out-of-domain (OOD) learning**, where strong in-domain performance **does not guarantee** successful generalization.

Next Lab: Find out how would distribution change impact model generalization

Test Your Self

- **Not independent:** Web-cam dataset may capture many similar input
- **Training and testing have different data distribution**: When there is distribution shift in real-world test data